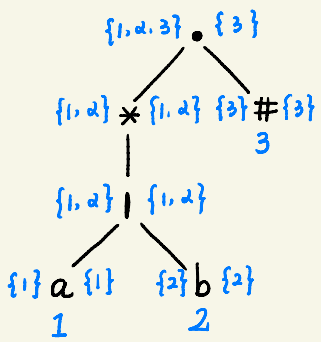


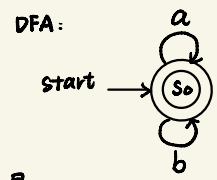
$$1) (a|b)^*$$

语法分析树



计算 followpos

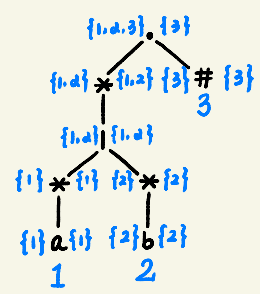
pos	followpos
1	{1, 2, 3}
2	{1, 2, 3}
3	Φ



最小化 DFA:
同上.

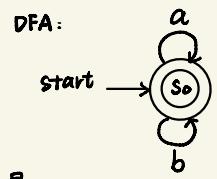
$$2) (a^* | b^*)^*$$

语法分析树



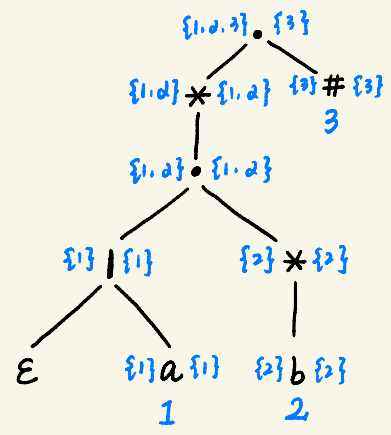
计算 followpos

pos	followpos
1	{1, 2, 3}
2	{1, 2, 3}
3	Φ



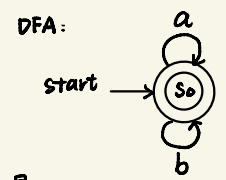
最小化 DFA:
同上.

$$c) ((\epsilon | a) b^*)^*$$



计算 followpos

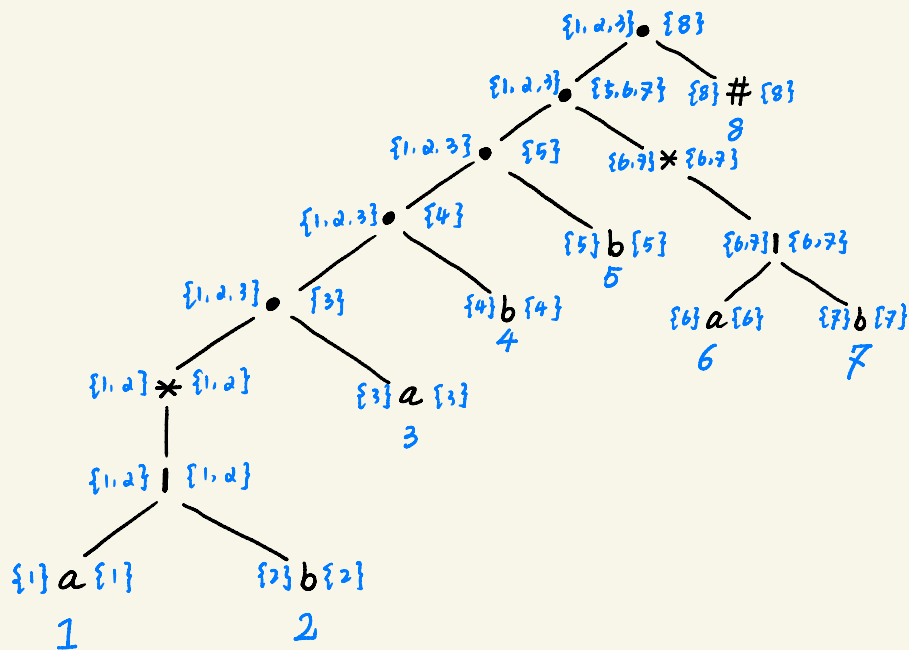
pos	followpos
1	{1, 2, 3}
2	{1, 2, 3}
3	Φ



最小化 DFA:
同上.

d) $(a|b)^*abb(a|b)^*$

语法分析树



计算 followpos

pos	followpos
1	{1, 2, 3}
2	{1, 2, 3}
3	{4}
4	{5}
5	{6, 7, 8}
6	{6, 7, 8}
7	{6, 7, 8}
8	Φ

(集)

(此处简写 followpos(i) 为 $f(i)$)

$$S_0 = \text{firstpos}(\text{root}) = \{1, 2, 3\} = S_0$$

↓ mark S_0

$$a: f(1) \cup f(3) = \{1, 2, 3, 4\} = S_1$$

$$\text{move}(S_0, a) = S_1$$

$$b: f(2) = S_0$$

$$\text{move}(S_0, b) = S_0$$

↓ mark S_1

$$a: f(1) \cup f(3) = S_1$$

$$\text{move}(S_1, a) = S_1$$

$$b: f(2) \cup f(4) = \{1, 2, 3, 5\} = S_2$$

$$\text{move}(S_1, b) = S_2$$

↓ mark S_2

$$a: f(1) \cup f(3) = S_1$$

$$\text{move}(S_2, a) = S_1$$

$$b: f(2) \cup f(5) = \{1, 2, 3, 6, 7, 8\} = S_3$$

$$\text{move}(S_2, b) = S_3$$

↓ mark S_3

$$a: f(1) \cup f(3) \cup f(6) = \{1, 2, 3, 4, 6, 7, 8\} = S_4$$

$$\text{move}(S_3, a) = S_4$$

$$b: f(2) \cup f(6) \cup f(7) = S_3$$

$$\text{move}(S_3, b) = S_3$$

↓ mark S_4

$$a: f(1) \cup f(3) \cup f(6) = S_4$$

$$\text{move}(S_4, a) = S_4$$

$$b: f(2) \cup f(4) \cup f(7) = \{1, 2, 3, 5, 6, 7, 8\} = S_5$$

$$\text{move}(S_4, b) = S_5$$

↓ mark S_5

$$a: f(1) \cup f(3) \cup f(6) = S_4$$

$$\text{move}(S_5, a) = S_4$$

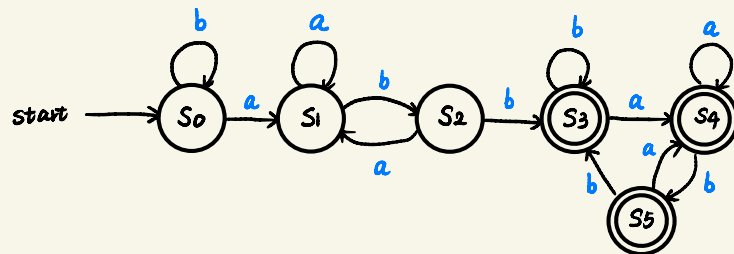
$$b: f(2) \cup f(5) \cup f(7) = S_3$$

$$\text{move}(S_5, b) = S_3$$

start state S_0

accepting states $\{S_3, S_4, S_5\}$

DFA:



最小 DFA:

